

- Consider another diagnosis such as a comorbid medical condition



Best Augmenting Combos for Partial Response or Treatment Resistance

- Benzodiazepines are frequently used as augmenting agents for antipsychotics and mood stabilizers in the treatment of psychotic and bipolar disorders
- Benzodiazepines are frequently used as augmenting agents for SSRIs and SNRIs in the treatment of anxiety disorders
- Not generally rational to combine with other benzodiazepines
- Caution if using as an anxiolytic concomitantly with other sedative hypnotics for sleep

Tests

- In patients with seizure disorders, concomitant medical illness, and/or those with multiple concomitant long-term medications, periodic liver tests and blood counts may be prudent

Side Effects

How Drug Causes Side Effects

- Same mechanism for side effects as for therapeutic effects –namely due to excessive actions at benzodiazepine receptors

- Long-term adaptations in benzodiazepine receptors may explain the development of dependence, tolerance, and withdrawal
- Side effects are generally immediate, but immediate side effects often disappear in time

Notable Side Effects

- * Sedation, fatigue, depression
- * Dizziness, ataxia, slurred speech, weakness
- * Forgetfulness, confusion
- * Hyperexcitability, nervousness
- * Pain at injection site
- Rare hallucinations, mania
- Rare hypotension
- Hypersalivation, dry mouth



Life-Threatening or Dangerous Side Effects

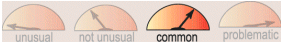
- Respiratory depression, especially when taken with CNS depressants in overdose
- Rare hepatic dysfunction, renal dysfunction, blood dyscrasias

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount
- Especially at initiation of treatment or when dose increases
- Tolerance often develops over time

What to Do About Side Effects

- Wait
- Wait
- Wait
- Lower the dose
- Take largest dose at bedtime to avoid sedative effects during the day
- Switch to another agent
- Administer flumazenil if side effects are severe or life-threatening

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- Oral: 2–6 mg/day in divided doses, largest dose at bedtime

- Injection: 4 mg administered slowly
- Catatonia: 1–2 mg per dose

Dosage Forms

- Tablet 0.5 mg, 1 mg, 2 mg
- Liquid 2 mg/mL
- Injection 2 mg/mL, 4 mg/mL

How to Dose

- Oral: initial 2–3 mg/day in 2–3 doses; increase as needed, starting with evening dose; maximum generally 10 mg/day
- Injection: initial 4 mg administered slowly; after 10–15 minutes may administer again
- Take liquid formulation with water, soda, applesauce, or pudding
- Catatonia: initial 1–2 mg; can repeat in 3 hours and then again in another 3 hours if necessary



Dosing Tips

- * One of the few benzodiazepines available in an oral liquid formulation
- * One of the few benzodiazepines available in an injectable formulation

- Lorazepam injection is intended for acute use; patients who require long-term treatment should be switched to the oral formulation
- Use lowest possible effective dose for the shortest possible period of time (a benzodiazepine-sparing strategy)
- Assess need for continued treatment regularly
- Risk of dependence may increase with dose and duration of treatment
- For interdose symptoms of anxiety, can either increase dose or maintain same total daily dose but divide into more frequent doses
- Can also use an as-needed occasional “top-up” dose for interdose anxiety
- Because panic disorder can require doses higher than 6 mg/day, the risk of dependence may be greater in these patients
- Some severely ill patients may require 10 mg/day or more
- Frequency of dosing in practice is often greater than predicted from half-life, as duration of biological activity is often shorter than pharmacokinetic terminal half-life

Overdose

- Fatalities can occur; hypotension, tiredness, ataxia, confusion, coma

Long-Term Use

- Evidence of efficacy up to 16 weeks

- Risk of dependence, particularly for treatment periods longer than 12 weeks and especially in patients with past or current polysubstance abuse

Habit Forming

- Lorazepam is a Schedule IV drug
- Patients may develop dependence and/or tolerance with long-term use

How to Stop

- Patients with history of seizure may seize upon withdrawal, especially if withdrawal is abrupt
- Taper by 0.5 mg every 3 days to reduce chances of withdrawal effects
- For difficult-to-taper cases, consider reducing dose much more slowly once reaching 3 mg/day, perhaps by as little as 0.25 mg per week or less
- For other patients with severe problems discontinuing a benzodiazepine, dosing may need to be tapered over many months (i.e., reduce dose by 1% every 3 days by crushing tablet and suspending or dissolving in 100 mL of fruit juice and then disposing of 1 mL and drinking the rest; 3–7 days later, dispose of 2 mL, and so on). This is both a form of very slow biological tapering and a form of behavioral desensitization

- Be sure to differentiate reemergence of symptoms requiring reinstitution of treatment from withdrawal symptoms
- Benzodiazepine-dependent anxiety patients and insulin-dependent diabetics are not addicted to their medications. When benzodiazepine-dependent patients stop their medication, disease symptoms can reemerge, disease symptoms can worsen (rebound), and/or withdrawal symptoms can emerge

Pharmacokinetics

- Elimination half-life 10–20 hours
- No active metabolites
- Food does not affect absorption



Drug Interactions

- Increased depressive effects when taken with other CNS depressants (see Warnings below)
- Valproate and probenecid may reduce clearance and raise plasma concentrations of lorazepam
- Oral contraceptives may increase clearance and lower plasma concentrations of lorazepam
- Flumazenil (used to reverse the effects of benzodiazepines) may precipitate seizures and should not be used in patients treated for seizure disorders with lorazepam

Other Warnings/Precautions

- Boxed warning regarding the increased risk of CNS depressant effects when benzodiazepines and opioid medications are used together, including specifically the risk of slowed or difficulty breathing and death
- If alternatives to the combined use of benzodiazepines and opioids are not available, clinicians should limit the dosage and duration of each drug to the minimum possible while still achieving therapeutic efficacy
- Patients and their caregivers should be warned to seek medical attention if unusual dizziness, lightheadedness, sedation, slowed or difficulty breathing, or unresponsiveness occur
- Dosage changes should be made in collaboration with prescriber
- Use with caution in patients with pulmonary disease; rare reports of death after initiation of benzodiazepines in patients with severe pulmonary impairment
- History of drug or alcohol abuse often creates greater risk for dependency
- Use oral formulation only with extreme caution if patient has obstructive sleep apnea; injection is contraindicated in patients with sleep apnea
- Some depressed patients may experience a worsening of suicidal ideation

- Some patients may exhibit abnormal thinking or behavioral changes similar to those caused by other CNS depressants (i.e., either depressant actions or disinhibiting actions)

Do Not Use

- If patient has angle-closure glaucoma
- If patient has sleep apnea (injection)
- Must not be given intra-arterially because it may cause arteriospasm and result in gangrene
- If there is a proven allergy to lorazepam or any benzodiazepine

Special Populations

Renal Impairment

- 1–2 mg/day in 2–3 doses

Hepatic Impairment

- 1–2 mg/day in 2–3 doses
- Because of its short half-life and inactive metabolites, lorazepam may be a preferred benzodiazepine in some patients with liver disease

Cardiac Impairment

- Benzodiazepines have been used to treat anxiety associated with acute myocardial infarction
- Rare reports of QTc prolongation in patients with underlying arrhythmia
- Lorazepam may be used as an adjunct to control drug-induced cardiovascular emergencies

Elderly

- 1–2 mg/day in 2–3 doses
- May be more sensitive to sedative or respiratory effects

Children and Adolescents

- Oral: safety and efficacy not established in children under age 12
- Injection: safety and efficacy not established in children under age 18
- Long-term effects of lorazepam in children/adolescents are unknown
- Should generally receive lower doses and be more closely monitored

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy

letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001

- Possible increased risk of birth defects when benzodiazepines are taken during pregnancy
- Because of the potential risks, lorazepam is not generally recommended as treatment for anxiety during pregnancy, especially during first trimester
- Drug should be tapered if discontinued
- Infants whose mothers received a benzodiazepine late in pregnancy may experience withdrawal effects
- Neonatal flaccidity has been reported in infants whose mothers took a benzodiazepine during pregnancy
- Seizures, even mild seizures, may cause harm to the embryo/fetus

Breast Feeding

- Some drug is found in mother's breast milk
- * Recommended either to discontinue drug or bottle feed
- Effects on infant have been observed and include feeding difficulties, sedation, and weight loss

The Art of Psychopharmacology

Potential Advantages

- Rapid onset of action
- Availability of oral liquid as well as injectable dosage formulations

Potential Disadvantages

- Euphoria may lead to abuse
- Abuse especially risky in past or present substance abusers
- Possibly more sedation than some other benzodiazepines commonly used to treat anxiety

Primary Target Symptoms

- Panic attacks
- Anxiety
- Muscle spasms
- Incidence of seizures (adjunct)

Pearls

* One of the most popular and useful benzodiazepines for treatment of agitation associated with psychosis, bipolar disorder, and other disorders, especially in the inpatient setting; this is due in part to useful sedative properties and flexibility of administration with oral tablets, oral liquid, or injectable formulations, which is often useful in treating uncooperative patients

- Is a very useful adjunct to SSRIs and SNRIs in the treatment of numerous anxiety disorders
 - Though not systematically studied, benzodiazepines, and lorazepam in particular, have been used effectively to treat catatonia and are the initial recommended treatment
 - Not effective for treating psychosis as a monotherapy, but can be used as an adjunct to antipsychotics
 - Not effective for treating bipolar disorder as a monotherapy, but can be used as an adjunct to mood stabilizers and antipsychotics
 - Because of its short half-life and inactive metabolites, lorazepam may be preferred over some benzodiazepines for patients with liver disease
- * Lorazepam may be preferred over other benzodiazepines for the treatment of delirium
- When treating delirium, lorazepam is often combined with haloperidol, with the haloperidol dose 2 times the lorazepam dose
- * Lorazepam is often used to induce pre-operative anterograde amnesia to assist in anesthesiology
- May both cause depression and treat depression in different patients
 - Clinical duration of action may be shorter than plasma half-life, leading to dosing more frequently than 2–3 times daily in some patients

- When using to treat insomnia, remember that insomnia may be a symptom of some other primary disorder itself, and thus warrant evaluation for comorbid psychiatric and/or medical conditions

Suggested Reading

Bonnet MH , Arand DL . The use of lorazepam TID for chronic insomnia .
Int Clin Psychopharmacol 1999 ;**14** :81 –9.

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Greenblatt DJ . Clinical pharmacokinetics of oxazepam and lorazepam .
Clin Pharmacokinet 1981 ;**6** :89 –105.

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Starreveld E , Starreveld AA . Status epilepticus. Current concepts and
management . *Can Fam Physician* 2000 ;**46** :1817 –23.

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Wagner BK , O'Hara DA , Hammond JS . Drugs for amnesia in the ICU .
Am J Crit Care 1997 ;**6** :192 –201 .

Loxapine

Therapeutics

Brands

- Loxitane
- Adasuve (Staccato loxapine, inhaled loxapine)
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: dopamine and serotonin receptor antagonist (DS-RAn)
- Conventional antipsychotic (neuroleptic, dopamine 2 antagonist, serotonin dopamine antagonist)

Commonly Prescribed for

(bold for FDA approved)

- **Schizophrenia**
- **Acute treatment of agitation associated with schizophrenia or bipolar disorder**

- Other psychotic disorders
- Bipolar disorder

How the Drug Works

- Blocks dopamine 2 receptors, reducing positive symptoms of psychosis
- * Although classified as a conventional antipsychotic, loxapine is a potent serotonin 2A antagonist
- Serotonin 2A antagonist properties might be relevant at low doses, but generally are overwhelmed by high dosing

How Long Until It Works

- Psychotic symptoms can improve within 1 week, but it may take several weeks for full effect on behavior

If It Works

- Most often reduces positive symptoms in schizophrenia but does not eliminate them
- Most schizophrenic patients do not have a total remission of symptoms but rather a reduction of symptoms by about a third
- Continue treatment in schizophrenia until reaching a plateau of improvement
- After reaching a satisfactory plateau, continue treatment for at least a year after first episode of psychosis in schizophrenia

- For second and subsequent episodes of psychosis in schizophrenia, treatment may need to be indefinite
- Reduces symptoms of acute psychotic mania but not proven as a mood stabilizer or as an effective maintenance treatment in bipolar disorder
- After reducing acute psychotic symptoms in mania, switch to a mood stabilizer and/or an atypical antipsychotic for mood stabilization and maintenance

If It Doesn't Work

- Consider trying one of the first-line atypical antipsychotics
- Consider trying another conventional antipsychotic
- If 2 or more antipsychotic monotherapies do not work, consider clozapine



Best Augmenting Combos for Partial Response or Treatment Resistance

- Augmentation of conventional antipsychotics has not been systematically studied
- Addition of a mood-stabilizing anticonvulsant such as valproate, carbamazepine, or lamotrigine may be helpful in both schizophrenia and bipolar mania
- Augmentation with lithium in bipolar mania may be helpful
- Addition of a benzodiazepine, especially short-term for agitation

Tests

* Since conventional antipsychotics are frequently associated with weight gain, before starting treatment, weigh all patients and determine if the patient is already overweight (BMI 25.0–29.9) or obese (BMI >30)

- Before giving a drug that can cause weight gain to an overweight or obese patient, consider determining whether the patient already has pre-diabetes (fasting plasma glucose 100–125 mg/dL), diabetes (fasting plasma glucose >126 mg/dL), or dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol), and treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

* Monitor weight and BMI during treatment

* Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics

* While giving a drug to a patient who has gained >5% of initial weight, consider evaluating for the presence of pre-diabetes, diabetes, or dyslipidemia, or consider switching to a different antipsychotic

- Should check blood pressure in the elderly before starting and for the first few weeks of treatment
- Monitoring elevated prolactin levels of dubious clinical benefit

- Patients with low white blood cell count (WBC) or history of drug-induced leukopenia/neutropenia should have complete blood count (CBC) monitored frequently during the first few months and loxapine should be discontinued at the first sign of decline of WBC in the absence of other causative factors
- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)

Side Effects

How Drug Causes Side Effects

- Acute blockade of dopamine 2 receptors in the striatum can cause drug-induced parkinsonism, dystonia, or akathisia
- Chronic blockade of dopamine 2 receptors in the striatum can cause tardive dyskinesia
- By blocking dopamine 2 receptors in the pituitary, it can cause elevations in prolactin
- By blocking dopamine 2 receptors excessively in the mesocortical and mesolimbic dopamine pathways, especially at high doses, it can cause worsening of negative and cognitive symptoms (neuroleptic-induced deficit syndrome)
- Blocking muscarinic cholinergic receptors can cause dry mouth, blurred vision, urinary retention, constipation, and paralytic ileus

- Antihistaminic actions may cause sedation, weight gain
- Blocking alpha 1 adrenergic receptors can cause dizziness, hypotension, and syncope
- Mechanism of weight gain and any possible increased incidence of diabetes or dyslipidemia with conventional antipsychotics is unknown

Notable Side Effects

- * Neuroleptic-induced deficit syndrome
- * Akathisia
- * Drug-induced parkinsonism
- * Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration
- * Rare tachycardia
- * Galactorrhea, amenorrhea
- Sedation
- Dry mouth, constipation, vision disturbance, urinary retention
- Hypotension, tachycardia



Life-Threatening or Dangerous Side Effects

- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability

with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure

- Rare agranulocytosis
- Rare hepatocellular injury
- Rare seizures
- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis
- Bronchospasm, with the potential to lead to respiratory distress and respiratory arrest (inhalant)

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount
- Sedation is usually transient
- Sedation is usually dose-dependent and may not be experienced at low doses where loxapine may function as an atypical antipsychotic (e.g., <50 mg/day; especially 5–25 mg/day)

What to Do About Side Effects

- Wait
- Wait
- Wait
- For drug-induced parkinsonism, add an anticholinergic agent
- Beta blockers, benzodiazepines, or 5HT2A antagonists (e.g., mirtazapine, cyproheptadine) may reduce akathisia
- Reduce the dose
- For sedation, give at night
- Switch to an atypical antipsychotic
- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia
- Metformin may help prevent or reverse antipsychotic-induced weight gain

Best Augmenting Agents for Side Effects

- Benztropine, trihexyphenidyl, or amantadine for drug-induced parkinsonism
- Beta blockers, benzodiazepines, or 5HT2A antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 60–100 mg/day in divided doses

Dosage Forms

- Capsule 6.8 mg loxapine succinate equivalent to 5 mg loxapine, 13.6 mg loxapine succinate equivalent to 10 mg loxapine, 34.0 mg loxapine succinate equivalent to 25 mg loxapine, 68.1 mg loxapine succinate equivalent to 50 mg loxapine
- Oral liquid 25 mg/mL (discontinued in USA)
- Injection 50 mg/mL (discontinued in USA)
- Inhalant 10 mg unit in a single-use inhaler

How to Dose

- Initial 20 mg/day in 2 doses; titrate over 7–10 days to 60–100 mg/day in 2–4 doses; maximum generally 250 mg/day
- Take liquid formulation in orange or grapefruit juice
- Inhalation powder: 10 mg by oral inhalation using an inhaler; only 1 dose per 24 hours; must be administered by a healthcare professional in a setting with immediate onsite access to equipment and personnel trained to manage acute bronchospasm, including advanced airway management

Dosing Tips

- Has conventional antipsychotic properties at originally recommended doses (i.e., starting at 10 mg twice a day, maintenance 60–100 mg/day, maximum 250 mg/day given in 2 divided doses)
- * Binding studies, PET studies, and anecdotal clinical observations suggest that loxapine may be atypical at lower doses (perhaps 5–30 mg/day) but further studies needed
- Anecdotal evidence that many patients can be maintained at 20–60 mg/day as monotherapy
- To augment partial responders to an atypical antipsychotic, consider doses of loxapine as low as 5–60 mg/day, but use full doses if necessary
- No formal studies, but some patients may do well on once daily dosing, especially at night, rather than twice daily dosing
- Available as 5 mg and 10 mg capsules for low-dose use and as 25 mg and 50 mg capsules for routine use
- Available as a liquid dosage formulation
- Available for acute inhalation administration
- Prior to administering inhalation powder, screen all patients for a history of pulmonary disease, and examine patients (including chest auscultation) for respiratory abnormalities (e.g., wheezing)
- After administering inhalation powder, monitor patients for signs and symptoms of bronchospasm at least every 15 minutes for at least an hour

- Available for acute intramuscular administration (50 mg/mL)
- Intramuscular loxapine may have faster onset of action and superior efficacy for agitated/excited and aggressive behavior in some patients than intramuscular haloperidol
- In the acute situation, give 25–50 mg intramuscularly (0.5–1.0 mL of 50 mg/mL solution) with onset of action within 60 minutes
- When initiating therapy with an atypical antipsychotic in an acute situation, consider short-term intramuscular loxapine to “lead in” to orally administered atypical; e.g., initiate oral dosing of an atypical antipsychotic with 25–50 mg loxapine 2–3 times a day intramuscularly to achieve antipsychotic effects without drug-induced parkinsonism and sedation
- When using loxapine to “top up” previously stabilized patients now decompensating, may use loxapine as single 25–50 mg doses as needed intramuscularly or as oral liquid or tablets
- Patients receiving atypical antipsychotics may occasionally require a “top-up” of a conventional antipsychotic to control aggression or violent behavior
- Treatment should be suspended if absolute neutrophil count falls below $1000/\text{mm}^3$

Overdose

- Deaths have occurred; drug-induced parkinsonism, CNS depression, cardiovascular effects, hypotension, seizures, respiratory depression,